AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 4 Submission; due Tuesday, November 10, 2026

Name: Vlad Paun

Student number: 21291453

[1] Let $A \subseteq \mathbb{R}^n$ be a non-empty subset, and let $\mathrm{Map}(A, \mathbb{R}^m)$ be the space of maps from A to $\mathbb{R}^m$.

[a] Let (f_k) be a sequence in $\mathrm{Map}(A, \mathbb{R}^m)$ which converges uniformly to a *bounded* function f, so $f \in B(A, \mathbb{R}^m)$. Prove that the f_k are *eventually bounded*. That is, prove that there exists $N \in \mathbb{N}$, such that $k \geq N$ implies $f_k \in B(A, \mathbb{R}^m)$.

Answer:

Proof. Since f_k converges uniformly to f:

$$\forall \varepsilon > 0 : \ \exists N \in \mathbb{N} : \ \forall k \geq N : \ \|f_k(x) - f(x)\| < \varepsilon \quad \forall x \in A$$

So take $\varepsilon = 1$ and the respective N. Also, since f is bounded, take M such that $\forall x \in A : \ \|f(x)\| \leq M$
Now,

$$\begin{aligned} \|f_k(x)\| &= \|f_k(x) - f(x) + f(x)\| && \forall x \in A \\ &\leq \|f_k - f(x)\| + \|f(x)\| && \forall x \in A \\ &< \varepsilon + M = 1 + M && \forall k \geq N, \forall x \in A \end{aligned}$$

Therefore, $\forall k \geq N : \ f_k \in B(A, \mathbb{R}^m)$ □

[b] Suppose (f_k) and (g_k) are two sequences in $\mathrm{Map}(A, \mathbb{R}^n)$ which converge uniformly to *bounded* functions $f, g \in B(A, \mathbb{R}^m)$. Prove that the sequence $(\langle f_k, g_k \rangle)_{k \in \mathbb{N}}$ in $\mathrm{Map}(A, \mathbb{R})$ converges uniformly to $f \cdot g \in \mathrm{Map}(A, \mathbb{R})$, where $\langle \cdot, \cdot \rangle$ is the dot product in $\mathbb{R}^m$.

Answer:

Proof. We have to show

$$\forall \varepsilon > 0 : \ \exists N \in \mathbb{N} : \ \forall k \geq N : \ |\langle f_k, g_k \rangle (x) - \langle f, g \rangle (x)| < \varepsilon \quad \forall x \in A$$

By 1[a], g_k is eventually bounded, so let N_0 such that $\forall k \geq N_0 : \ \|g_k(x)\| \leq M_0$.
Let M_f, M_g be the bounds of f, g.
Let arbitrary $\varepsilon > 0$. Let $\varepsilon' = \frac{\varepsilon}{M_0 + M_f}$. Let N_f, N_g such that:

$$\forall k \geq N_f : \ \|f_k(x) - f(x)\| < \varepsilon' \quad \forall x \in A$$

$$\forall k \geq N_g : \ \|g_k(x) - g(x)\| < \varepsilon' \quad \forall x \in A$$

Take $N = \max(N_f, N_g, N_0)$ and take $k \geq N$.
Now,

$$\begin{aligned} |\langle f_k, g_k \rangle (x) - \langle f, g \rangle (x)| &= |\langle f_k, g_k \rangle (x) - \langle f, g_k \rangle (x) + \langle f, g_k \rangle (x) - \langle f, g \rangle (x)| \\ &= |\langle f_k - f, g_k \rangle (x) + \langle f, g_k - g \rangle (x)| \\ &\leq |\langle f_k - f, g_k \rangle (x)| + |\langle f, g_k - g \rangle (x)| \\ &\leq \|(f_k(x) - f(x))\| \|g_k(x)\| + \|f(x)\| \|(g_k(x) - g(x))\| \\ &< \varepsilon' \|g_k(x)\| + \varepsilon' \|f(x)\| \\ &< \varepsilon'(M_0 + M_f) = \frac{\varepsilon}{M_0 + M_f}(M_0 + M_f) = \varepsilon \end{aligned}$$

□

[c] Show that the assumption of boundedness of the limits in part [b] is essential as follows. Show that the sequence (f_k) in $M(\mathbb{R},\mathbb{R})$ given by $f_k(x) = x + \frac{1}{k}$ converges uniformly on $\mathbb{R}$ to the unbounded function $f(x) = x$, but that $f_k^2 \in M(\mathbb{R},\mathbb{R})$ does not converge uniformly to f^2.

Answer:

Proof. Let arbitrary $\varepsilon > 0$. Take $N > \frac{2}{\varepsilon}$

$$\begin{aligned} |f_k(x) - f(x)| &= \left| x + \frac{1}{k} - x \right| = \left| \frac{1}{k} \right| \\ &\leq \frac{1}{N} < \frac{\varepsilon}{2} < \varepsilon \qquad\qquad \forall k \geq N, \forall x \in \mathbb{R} \end{aligned}$$

So f_k converges uniformly to f.

Now, We have to show

$$\exists \varepsilon > 0 : \ \forall N \in \mathbb{N} : \ \exists k \geq N : \ \exists x \in \mathbb{R} : \ \left| f_k^2(x) - f^2(x) \right| \geq \varepsilon$$

Take $\varepsilon = 1$. Let arbitrary $N \in \mathbb{N}$. Take $k = N$. Take $x = k$

$$\begin{aligned} \left| f_k^2(x) - f^2(x) \right| &= \left| x^2 + \frac{2x}{k} + \frac{1}{k^2} - x^2 \right| = \left| \frac{2x}{k} + \frac{1}{k^2} \right| \\ &= \left| 2 + \frac{1}{N^2} \right| > 2 > 1 = \varepsilon \end{aligned}$$

So f_k^2 does not converge uniformly to f^2. □

[2] Let $f \in \mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ be a map, and suppose that f is *continuous* at $\mathbf{x}_0 \in \mathbb{R}^n$.

Define a sequence (f_k) in $\mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ by $f_k(\mathbf{x}) = f(\mathbf{x}_0 + \frac{1}{k}\mathbf{x})$. Let $g \in \mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ be the *constant map* $g(\mathbf{x}) = f(\mathbf{x}_0)$ for all $\mathbf{x} \in \mathbb{R}^n$.

Prove that, for any $R > 0$, the sequence (f_k) converges uniformly to g on the set $B_R(\mathbf{0})$.

Answer:

[3] For $k \in \mathbb{N}$, define $f_k : [-1, 1] \to \mathbb{R}$ by $f_k(x) = (1 - x^2)^k$.

[a] For any $\delta \in (0, 1)$, prove that the sequence of functions (f_k) *converge uniformly* on the subset $[-1, -\delta] \cup [\delta, 1]$ of $\mathbb{R}$.

Answer:

[b] Prove that the sequence of functions (f_k) *does not converge uniformly* on the subset $[-1, 1]$ of $\mathbb{R}$.

Answer:

[4] Let $A \subseteq \mathbb{R}^n$ be non-empty. Let (f_k) be a sequence in $\mathrm{Map}(A, \mathbb{R}^m)$ which converges *uniformly* to a limit $f \in \mathrm{Map}(A, \mathbb{R}^m)$. We proved in class that if every f_k is continuous on A, then f is also continuous on A.

[a] Adapt that proof to show that if every f_k is *uniformly* continuous on A, then f is also *uniformly* continuous on A.

Answer:

[b] Explain why we do not need to do any extra work if A is compact. That is, why does part [a] follow immediately if A is compact?

Answer:

[5] COMING SOON

Answer:

[6] COMING SOON

Answer:

[7] COMING SOON

Answer:

[8] COMING SOON

Answer:

[9] COMING SOON

Answer:

[10] COMING SOON

Answer: